Name

Physical and chemical changes 

What is a physical change?

_____

What is a chemical change?

_____

Identify each of the following as a physical or a chemical change. Then, explain why each is a chemical or physical change.

1. You leave your bicycle out in the rain and it rusts_____
2. A sugar cube dissolved in water_____
3. Pure potassium is normally stored in oil. When the oil is removed, a flame is produced because the potassium can now react with oxygen in the air_____
4. Sand is mixed with water_____
5. Scientists break up water (called hydrolysis) by separating it into 02 and H2_____
6. You are cleaning your bathroom and you accidentally mix bleach and ammonia, which produces a toxic gas, which makes you pass out_____
7. Burning coal for a barbecue

   _____

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_____
_____

8. Chewing up a bite of hamburger_____
_____
_____

9. You take a shower and your wet hair begins to dry_____
_____
_____

10. Silver nitrate and sodium chloride mix to form a grey-violet precipitate
_____
_____
_____

11. Trimming a plant because it has grown too high
_____
_____
_____

12. Mashing up potatoes to make mashed potatoes
_____
_____
_____

13. Hydrogen peroxide is poured on some liver, the liver begins to break down
_____
_____
_____

Write your own example below and identify it as a chemical or physical change and explain why.

_____
_____
_____

Identify each statement as true of false

a. A change in size or shape is a physical change

b. A chemical change means a new substance with new properties was formed

c. An example of a chemical change is when water freezes

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d. When platinum is heated, then cooled to its original state, we say this is a physical change.

e. When milk turns sour, this is a physical change because a change in odour does not indicate a chemical change.

f. When magnesium is burned, ash forms. We say this is a physical change because the magnesium looks different.

g. When citric acid and baking soda mix, carbon dioxide is produced and the temperature decreases. This must be a chemical change.

Signs that a physical or chemical change has occurred. Fill in the spaces (size, colour, light, odour, sound, state, bubbles, heat, shape)

Physical change	Chemical change
A change in S_____	A change in C_____
A change in S_____	B_____ given off
A change in S_____	O_____ given off
	H_____ given off
	L_____ given off
	S_____ given off
	A P_____ is formed

TEACHER DEMONSTRATIONS



1. Burning a match

Record as many observations as you can_____

_____

Was this a chemical or physical change? Why?

_____

_____

2. *Ripping a piece of paper into small pieces*

Name

Record as many observations as you can_____

_____

Was this a chemical or physical change? Why?

_____

_____

3. Ice left in a container at room temperature

Record as many observations as you can_____

_____

Was this a chemical or physical change? Why?

_____

_____

4. Coke bottle rocket

Record as many observations as you can_____

_____

Was this a chemical or physical change? Why?

_____

_____

5. Hydrogen balloon

Record as many observations as you can_____

_____

Was this a chemical or physical change? Why?

_____

_____

6. Mixing solutions – Potassium iodine with lead nitrate

Record as many observations as you can_____

_____

Was this a chemical or physical change? Why?

_____

_____

Which of the changes demonstrated by the teacher were

Physical? _____ Chemical? _____

_____ _____

Name _____ _____